

20/01/2025

(XI) - Power Set :- A power set is defined as the set or group of all subsets for any given set, including the empty set, which is denoted by $\{\}$ or Φ . A set that has ' n ' elements has 2^n subsets in all.

Example :- let Set $A = \{1, 2, 3\}$, total number of element in the set is 3. Therefore, there are 2^3 elements in the power set.

Subsets of Set A = $\{\}, \Phi, \{1\}, \{2\}, \{3\}, \{1, 2\}, \{2, 3\}, \{1, 3\}, \{1, 2, 3\}$

Power Set $P(A)$ = $\{\}, \{1\}, \{2\}, \{3\}, \{1, 2\}, \{2, 3\}, \{1, 3\}, \{1, 2, 3\}$

(XII) - Equivalent Set :- If the number of elements is the same for two different sets, then they are called **Equivalent Sets**. The order of sets does not matter. It is represented as :- $n(A) = n(B)$, where A & B are two different set with the same no of elements.

Example :- If $A = \{1, 2, 3, 4\}$ and $B = \{\text{Red, Blue, Black, Green}\}$.

* Operations on Sets

- (i) Union of Sets : Union of two sets A & B is represented by ' $A \cup B$ ' and it is a set which contains elements of both sets.

Example : $A = \{1, 2, 3, 4\}$, $B = \{2, 3, 5\}$

then

$$A \cup B = \{1, 2, 3, 4\} \cup \{2, 3, 5\}$$

$$A \cup B = \{1, 2, 3, 4, 5\},$$

- (ii) Intersection of Sets : Intersection of two sets A & B is represented by ' $A \cap B$ ' and it is a set which contains common elements of both sets.

Example : $A = \{1, 2, 3\}$, $B = \{2, 3, 5\}$

then

$$A \cap B = \{1, 2, 3\} \cap \{2, 3, 5\}$$

$$A \cap B = \{2, 3\},$$

- (iii) Difference of Sets : Difference of two sets A & B is represented by ' $A - B$ ' and it is a set which contains those elements of first set that are not present in second set.

Example : $A = \{1, 2, 3, 4\}$, $B = \{2, 3, 5\}$

$$A - B = \{1, 2, 3, 4\} - \{2, 3, 5\}$$

$$A - B = \{1, 4\},$$

(iv) Compliments of Sets :- Compliments of a set A is represented by " A^c " or A' and it is a set which contains those elements of universal set that are not present in set A .

Example : If $A = \{1, 2, 3\}$, $U = \{1, 2, 3, 4, 5\}$

then

$$[A^c = U - A]$$

$$A^c = \{1, 2, 3, 4, 5\} - \{1, 2, 3\}$$

$$A^c = \{4, 5\},$$

*

Properties of sets

Sets are represented as the combination of objects whose elements are fixed and can not be changed. The property of sets use a concept that is similar to the properties of basic operations on the number system.

- (i) Commutative Property:- For any two given sets A and B, the commutative property is represented as:-

$$A \cup B = B \cup A \quad \text{and} \quad A \cap B = B \cap A$$

Example:- Let $A = \{5, 6, 7\}$, $B = \{2, 4, 6, 8\}$

$$A \cup B = B \cup A$$

$$\{5, 6, 7\} \cup \{2, 4, 6, 8\} = \{2, 4, 6, 8\} \cup \{5, 6, 7\}$$

$$\{2, 4, 5, 6, 7, 8\} = \{2, 4, 5, 6, 7, 8\} //$$

or

$$A \cap B = B \cap A$$

$$\{5, 6, 7\} \cap \{2, 4, 6, 8\} = \{2, 4, 6, 8\} \cap \{5, 6, 7\}$$

$$\{6\} = \{6\} //$$

- (ii) Associative Property:- For the given 3 sets A, B and C the associative property is represented as:-

$$(A \cup B) \cup C = A \cup (B \cup C) \quad \text{and} \quad (A \cap B) \cap C = A \cap (B \cap C)$$

Example: Let $A = \{a, h, t\}$, $B = \{t, a, p\}$, $C = \{s, a, p\}$

$$(A \cup B) \cup C = A \cup (B \cup C)$$

$$(\{a, h, t\} \cup \{t, a, p\}) \cup \{s, a, p\} = \{a, h, t\} \cup (\{t, a, p\} \cup \{s, a, p\})$$

$$\{p, a, h, t\} \cup \{s, a, p\} = \{a, h, t\} \cup \{p, a, s, t\}$$

$$\{p, a, h, s, t\} = \{p, a, h, t, s\},$$

$$(A \cap B) \cap C = A \cap (B \cap C)$$

$$\{a, t\} \cap \{s, a, p\} = \{a, h, t\} \cap \{a, p\}$$

$$\{a\} = \{a\},$$

(iii) Distributive Property: For the given 3 sets A, B & C the

distributive Property are represented as:

$$A \cap (B \cup C) = (A \cap B) \cup (A \cap C) \text{ \& }$$

$$A \cup (B \cap C) = (A \cup B) \cap (A \cup C)$$

(iv) De-morgan's Law:

$$(A \cup B)' = A' \cap B'$$

$$(A \cap B)' = A' \cup B'$$

(v) Identity Property:-

$$A \cup \phi = A \quad \& \quad A \cap \underset{\text{set}}{U} = A$$

Example:-

$$\text{Let } A = \{3, 7, 11\}$$

$$B = \{x : x \text{ is a natural no less than } 0\}$$

$$A \cup B$$

$$\{3, 7, 11\} \cup \phi$$

$$= \{3, 7, 11\},$$

(vi) Complement Property:-

$$A \cup A' = U$$

Example:-

$$\text{If } U = \{1, 2, 3, 4, 5\} \quad \& \quad A = \{1, 2, 3\}$$

$$A' = U - A$$

$$= \{1, 2, 3, 4, 5\} - \{1, 2, 3\}$$

$$= \{4, 5\},$$

vii) Idempotent Property:-

$$A \cup A = A$$

$$A \cap A = A$$